NANDAN HEGDE

Senior Software Engineer | AI Engineer | Generative AI and Full-Stack Development

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linkedin.com/in/nandanhegde1096 | nandanhegde1.github.io/portfolio | github.com/Nandanhegde1 | Indian national, open to relocation

SUMMARY

Senior Software Engineer with 6+ years building production software. Led the AI workstream for about three years on a recruiting platform used by 10,000+ people, with a team of about 10 engineers and direct client contact on scope. Took four Generative AI (GenAI) features from design into production: an AI voice interviewer, generated interview flows, transcript fit scoring and a candidate ranking engine built on hybrid retrieval.

Six years hands-on in TypeScript, Angular and Node.js, including a zero-downtime migration of a 7-year-old AngularJS platform to Angular 19. Used Azure OpenAI in production. Side projects use Claude, Gemini and Azure AI Foundry: an MCP-based agent and a retrieval system checked by a CI evaluation harness.

TECHNICAL SKILLS

AI & LLM: Generative AI · large language models (LLMs) · AI agents and agentic workflows · LLM tool calling and function calling · Model Context Protocol (MCP) · prompt engineering and structured output · NLP and LLM-based extraction · AI guardrails and prompt injection handling · Azure OpenAI, Azure AI Foundry, Claude (Anthropic), Google Gemini

Retrieval & evaluation: retrieval-augmented generation (RAG) · BM25, embeddings and hybrid retrieval · vector search · vector databases (Qdrant, Milvus evaluated) · evaluation harness in side projects: gold sets, recall@k, CI gates

Languages & frameworks: TypeScript · JavaScript · Angular 2 to 19 (signals, standalone components, RxJS) · Node.js · Express · React · Python · FastAPI · Pydantic · SQL · HTML, CSS, SCSS · REST API design · server-sent events (SSE)

Cloud & DevOps: Azure, App Service, Azure Functions, Azure DevOps · Cloudflare Workers · Docker · Git, GitHub Actions, CI/CD · AWS, GCP, Terraform, Jenkins (2020–2022)

Observability & practice: LLMOps and observability with tracing (Langfuse), spend and latency ceilings and uptime monitoring · Microsoft SSO · PostgreSQL, MongoDB, Supabase · high-level and low-level design (HLD, LLD) · unit testing and branch coverage · Claude Code, GitHub Copilot · Agile, Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer

thinkbridge | Bengaluru, India | Mar 2022 – Present

Client: a US recruiting-technology firm. Led its AI workstream for about three years, from design through production, with a team of about 10 engineers. Owned design and code review, mentored the team through both and worked directly with the client on scope.

- **AI voice interviewer:** designed the interview flow, the conversation pathways on Bland AI and the fit-scoring model for an LLM voice agent that interviews each candidate in their own language. Demoed to the client at an industry tradeshow in Austin, US.
- **Fit scoring:** each interview transcript was scored against weighted criteria (skills, experience, location, availability, tenure) through schema-validated LLM extraction and knockout rules, with interview evidence weighted over the résumé.
- **Candidate ranking engine (in production):** led the design and review of the retrieval service that finds the strongest candidates for a job order. Hybrid retrieval with dense embeddings and BM25 in Qdrant, reranking, then a deterministic rule-based final score with weights recruiters can tune. Chose Qdrant after evaluating Milvus and set the model-cost strategy.
- **Platform modernization:** led the zero-downtime migration of a 7-year-old AngularJS recruiting platform to Angular 19, cutting initial load by about 30% (measured before and after). Integrated Power BI, Microsoft SSO and Bland AI, and shipped 15+ features across five sprints.
- **AI-assisted development:** drove Claude adoption across 30+ engineers and authored the team's guidelines for AI in the codebase, including review gates and what may not be delegated. Drove branch-coverage targets on critical modules, tracked in the Azure DevOps CI pipeline.
- **Service integration:** owned the REST contract and layering between the Python (FastAPI) machine-learning microservice and the platform's .NET backend, and reviewed that backend code weekly.
- **Recruiting tools:** built a browser extension that automated recruiting data capture for 500+ users, and a real-time candidate-data sync engine in Angular.

Senior System Engineer

Infosys | Bengaluru, India | Jan 2020 – Mar 2022

Promoted from System Engineer in January 2022. Clients: Swiss Re (global reinsurance) and Infosys internal platform engineering.

- **Corflow Claims, Swiss Re:** modernized the UI of a global insurance-claims platform serving 40+ countries (Angular, lazy loading, code splitting), with unit and end-to-end tests for critical components.
- **Cloud Migration Accelerator, internal:** built the UI and Node.js backend for a tool that generates boilerplate and infrastructure across Azure, AWS and Google Cloud (GCP) with Terraform and Jenkins CI/CD.

PERSONAL PROJECTS

Planning Desk | chat assistant that answers planning questions through MCP tools

Python, FastAPI, MCP, JSON-RPC, SSE | built on Azure AI Foundry, deployed on Gemini | planning-desk.onrender.com | github.com/Nandanhegde1/planning-desk

- Three MCP servers (feasibility, foreign exchange, discovery) run as long-lived processes speaking JSON-RPC over stdio, behind an MCP host that discovers their tools and dispatches calls. Progress and per-tool timings stream to the browser over server-sent events (SSE).
- Custom single-agent loop instead of LangGraph or LangChain, to control state directly, with hard ceilings on steps (8), tool calls (12), tokens (60,000) and time. When a tool fails, the agent reports the failure and does not make up an answer.
- Tool output is treated as data, never as instructions, which guards against indirect prompt injection. The model cannot overturn the rules engine's verdict. 180+ automated tests cover the orchestration and run without a live model or API key.

Ask GovAI | question answering over 700+ U.S. federal AI/ML contracts

TypeScript, BM25, query routing, CI evaluation harness | govai-contracts.nandanhegde1096.workers.dev/ask/

- Retrieval pipeline with no framework: a query router and BM25 search feed grounded, cited answers, and the model declines rather than invents when the data has no answer. Works with Claude or Gemini.
- Diagnosed the structural misses that keyword search cannot answer, such as superlatives and code filters, and routed them, raising recall@8 from 69% (9 of 13 queries) to 92% (12 of 13). Measured in-sample on a small gold set frozen before the router was built. The evaluation harness runs in CI and fails the build on a recall regression.

MCP Directory | automated index of 1,600+ MCP servers

TypeScript, GitHub Actions, Cloudflare Workers | mcp-directory.nandanhegde1096.workers.dev

- Unattended weekly pipeline that searches GitHub, removes duplicates, filters on metadata, classifies each server with a tested classifier and publishes only entries that pass automated checks. A cleanup pass then removed 162 false positives.

Portfolio (nandanhegde1.github.io/portfolio): Angular site with an animated replay of a recorded Planning Desk agent run, gated CI and 6-hourly uptime checks that open issues automatically. **IndiaTenders** (india-tenders.nandanhegde1096.workers.dev): search over Indian government tenders, with unit-tested parsers and automated checks that block bad data.

EDUCATION

B.E., Electronics and Telecommunication Engineering, Dayananda Sagar College of Engineering, Bengaluru, 2019